INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

RECHARGE/RECOVERY UNIT, FIRE EXTINGUISHER,

MONOBROMOTRIFLUOROMETHANE (CF_3Br), PNEUMATIC POWERED

This specification is approved for use within the USA Belvoir Research, Development , and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a monobromotrifluoromethane , recharge/recovery unit, operated with compressed air for charging and evacuating a CF_3Br fire extinguisher .

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Satndards (D0DISS) and supplement thereto, cited in the solicitation (see 6.2) .

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US ARMY AVIATION & TROOP CMD, ATTN AMSAT R EDS, ST Louis MO 63120-1798 by using the Standardization Document Impovement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4210

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SPECIFICATIONS

FEDERAL

L-P-535	Plastic Sheet (Sheeting) ; Plastic Strip: Poly (Vinyl Choride) and Poly (Vinyl Chloride-Vinyl Acetate) , Rigid.
QQ-P-35	Passivation Treatments for Corrosion-Resistant Steel .
TT-E-527	Enamel , Alkyd, Lusterless, Low Voc Content.
UU-C-190	Cardboard, Bristol (Drafting) .
ZZ-G-381	Gloves, Rubber, Industrial.
PPP-E-601	Boxes , Wood Cleated-Plywood.
PPP-B-621	Box , wood , Nailed and Lock-Corner.
PPP-B-636	Boxes , Shipping, Fiberboard .
PPP-T-60	Tape, packaging, Waterproof.

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-P-514	-	Plate, Identification, Instruction and Marking, Blank.
MIL-T-704	-	Treatment and Painting of Materiel.
MIL-H-775	-	Hose , Hose Assemblies; Rubber, Plastic, Fabric , or Metal (Including Tubing) ; and Associated Hardware: Packaging of.
MIL-M-12218	-	Monobromotrifluoromethane (Liquified) Technical Grade for Fire Extinguisher.
MIL-C-17711	-	Coatings, Chromate, For Zinc Alloy Castings and Hot-Dip Galvanized Surfaces.
MIL-C-52980	-	Compressor, Reciprocating and Rotary Gasoline-Engine-Driven, Air.

STANDARDS

FEDERAL

FED-STD-66		Steel, Chemical Composition and Hardenability.
FED-STD-595	-	Colors Used in Government Procurements.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-209		Military Standard Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment .
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-1366	-	Transportability Criteria.
MIL-STD-1434	-	Goggles (Metric) .

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MIL-STD-1472 -	Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1791	Designing for Internal Aerial Delivery in Fixed Wing Aircraft .
MS28889	Valve, Air, High Pressure Charging, 5000 PSI.

(Unless otherwise indicated, copies Of federal and military specifications, standards, and handbooks are available from: STDZN DCMNT ORDER DESK, BLDG 4D/ 700 ROBBINS AVE/ PHILADELPHIA PA 19111-5094.)

2.2 Non-Government Publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents rot listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2) .

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 153	Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
B 633	Electrodeposited Coating of Zinc on Iron and Steel .
D 3951	Standard Practices for Commercial Packaging.
D 3953	Standard Specification for Strapping, Flat Steel and Seals.

(Application for copies should be addressed to the AMERICAN SCTY & MTRLS , 1916 RACE STRET, PHILADELPHIA PA 19103.)

UNDERWRITERS LABORATORIES INC (UL)

UL 2083	Halon 1301 Recovery/Recyling Equipment .

(Application for copies should be addressed to: UNDERWRTRS LABS INC, 333 PFINGSTEN RD, NORTHBROOK IL 60062.)

2.3 Order of Precedence . In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 performance . The recharge/recovery unit shall be capable of filling 5-, 7-, and 14-pound fire extinguishers cylinders with monobromotrifluorornethane from a 150-pound halon supply cylinder The monobromotrifluoromethane (Halon 1301) shall conform to MIL-M-12218 and be supplied in a cylinder with a dip tube installed. The fire extinguisher cylinder shall be filled with the required

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amount of Halon 1301 and overpressurized with nitrogen to approximately 750 psi at 70 'F. The unit shall be capable of recovering the halon/nitrogen mixture, which may be overpressurized with nitrogen to 750 psi, and transferring it to a holding cylinder. The unit shall provide pressure reducing valves and pressure relief valves to ensure the safety of operating personnel . The pump shall be capable of drawing a minimum vacuum of 15 inches Of mercury on the fire extinguisher cylinder. The recharge or recovery procedure must be accomplished in 10 minutes or less. No venting of halon to the atmosphere is permitted during any of the operations. The recovery efficiency of the system shall be 99 percent minimum.

3.2 Description. The unit shall consist of a pump, pressure gauges, scale, thermometer, nitrogen regulator, toolbox, brackets or straps to retain the 150-pound nitrogen and 2 required halon supply cylinders ; a bracket to retain the fire extinguisher cylinder (see figure 3) in the horizontal position for charging, and in the vertical position (fire extinguisher cylinder inverted) for recovery; and all necessary piping, hoses, valves, adapters, quick disconnect fittings, and brackets required for functional completeness when charging a monobromotrifluoro-methane (Halon 1301) fire extinguisher from a 150-pound storage cylinder. The unit shall be capable of recovering a halon charge and transferring the halon to a storage cylinder. The system shall provide any pressure reducing or pressure relief features required to ensure safety of the operating personnel. Unless otherwise specified, the recovery cycle of the unit shall conform to the requirements of UL 2083 . Halon shall flow through the pump console to the fire extinguisher through a single hose. A ball valve shall be located in the hose at the fire extinguisher to control the flow of halon. Nitrogen shall flow directly from the nitrogen supply cylinder to the fire extinguisher and shall be controlled by a second ball valve in the hose at the fire extinguisher. The halon and nitrogen supply noses shall be joined in a tee at the fire extinguisher. The controls on the console shall be compound pressure gauge, inlet and outlet pressure gauges, pump, on-off; direction control valve, labeled, to fire extingisher, to recovery cylinder and a liquid sight glass/moisture indicator. The unit shall have provisions to preven~ the halon supply cylinder from being contaminated with nitrogen during charging and when a vacuum is pulled after recovery. A filter shall be provided which filters the halon during the recovery operation. The charging hose and the recovery hose shall have a quick disconnect fitting to mate with the charging adapter which shall mate with the MS28889-2 fill valve on the fire extinguisher. The unit shall be pneumatically operated by an air compressor conforming to MIL-C-52980, type 1, class optional, size 15 cfm at 175 psig. The unit shall have provisions for agitating the fire extinguisher cylinder during the overpressurization process to ensure complete mixture of the halon and nitrogen. The unit shall consist of a maximum of two pieces with maximum weight for any one item to be 195 pounds. Weights do not include halon or nitrogen supply cylinders. The unit shall be mounted on a metal base. Figures 1 and 2 are for illustration only and not intended to limit design. A pressure gauge and fittings shall be supplied for the

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halon recovery cylinder to monitor the pressure of the cylinder. A leak detector shall be supplied. The leak detector shall be a self-contained, battery operated unit and shall include carrying case . The detectcr shall have a minimum sensitivity of .4 oz of Halon 1301/yr. The detector shall operate in ambient temperature of 30 to 100 °F and shall require no calibration. A long, flexible corrosion resistant probe shall be supplied. The detector shall operate in highly contaminated atmospheres with no loss of sensitivity or contamination of sensor. Response time shall be 1 second or less and a continuous duty cycle is required. The detector shall weight no more than 24 ounces .

3.3 First article. Unless otherwise specified (see 6.2), a ‘sample shall be subjected to first article inspection (see 6.3) ,in accordance with 4 .3.

3.4 Material Material shall be as specified herein. Material *not* specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.4.1 Corrosion-resistant material. Where corrosion-resistant material is specified, any one of the following materials shall be used : corrosior. resistant steel (CRES) conforming to FED-STD-66, type 301 through 348, as applicable, which has been passivated in accordance with QQ-P-35 prior to assembly; copper; brass; bronze; beryllium-copper; copper-nickel alloys; or aluminum. Dissimilar metals, as defined and detailed in MIL-STD-885, shall net be used in intimate contact with each other unless protected by galvanic corrosion .

3.4.2 Corrosion-resistant treatment. Corrrosion-resisting treatment for steel shall consist of galvanizing in accordance with ASTM A153, to a G-90 coating designation, followed by a chromate conversion. coating in accordance with MIL-C-17721; or electrodeposited zinc coating in accordance with ASTM B 633, SC3 , type II or III as applicable. The selection of any treatment shall be in accordance with MIL-STD-889.

3.5 Environmental conditions . The units shall be capable of charging CF_3Br fire extinguisher cylinders at temperatures from +36 to 100 °F.

3.6 Safety. All moving parts, which are of such nature or so located as to be a hazard to operating or maintenance personnel, shall be fully enclosed or guarded. protective devices shall not impair the operating functions. Safety relief devices on pressure vessels such as supply and recovery cylinders and filter housings may vent to atmosphere; however, they may not vent to atmosphere during normal charging or recovery operations. Unless otherwise specified, the equipment shall meet the requirements Of UL 2083, parts subject to pressure, and pressure-limiting and pressure-relief devices. Non-functional sharp edges, projecting points, and excessive lengths of fastening devices shall be avoided.

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3.7 Hoses and valves. Hoses and valves, including all fittings necessary for adapting the supply cylinders and fire extinguisher cylinders to the inlet and outlet of the pump shall be furnished. Swivel-nut adapters shall be provided at both inlet and outlet ends of each hose and where required at valves that cannot be rotated in place for removal. The adapter at the fire extinguisher cylinder shall be a swivel fitting to mate with the MS28889-2 fill valve on the fire extinguisher cylinder, and have a quick disconnect fitting to mate with the charging and recovery hoses. All hoses shall be a minimum of 8 feet long from the exit point on the console to the ball valve on the end of the hose. The end of each hose shall be marked with its connection point, i.e. , halon supply, fire extinguisher, etc. All hose ends shall have protective caps and plugs . No hose connections shall protrude from the outside of the control console after all hoses have been disconnected.

3.8 Fittings. All fittings furnished shall be of corrosion-resistant material or steel, zinc plated, in accordance with ASTM B 633, SC3, type II or 111. All fittings subject to a pressure of 600 psi or greater during normal operation shall withstand an internal pressure of 1200 psig. The unit shall be supplied with male and female, 1/4-inch NPT quick disconnect fittings for making up the inlet air connection. Any fittings within the system that are disconnecting during the recharging or the recovery operation shall have an internal cap or plug to protect the fitting when not in use.

3.9 Air filter. The compressed air required to operate the pump shall be directed through a standard type air filter with a moisture trap . The filter shall be 40 microns or less and remove contaminants , and water vapor from the compressed air supply. The air filter shall have a means of viewing the level of the collected compressed air contaminants and a quick acting drain valve to discharge the contaminants from the filter housing. All housing and filter components shall be compatible with all currently expected sources of compressed air contamination.

3.10 Air Pressure regulator . A regulator shall be provided to regulate the supply air pressure to the required operating pressure. The regulator shall be rated for 300 psi maximum input, 0-125 psi adjustable output. The recommended operating pressure shall be aesigned to provide the proper pump performance. The regulator shall be installed between the air filter and pump control valve.

3.11 Recovery/ recharge pump. The recovery/recharge pump shall be of the pneumatically driven type. The pump shall have the ability to pump liquid halon, and when the liquid has been removed from the cylinder it can then pump the vapor to a minimum vacuum of 15 inches of mercury while providing a minimum discharge pressure of 300 psig. The pump speed shall be completely variable from zero to maximum halon flow. It shall be capable of starting against any discharge pressure load without the need of external unloaders or bypass valving. The pump shall be capable of dry operation for short durations without. adverse affects. The pump shall not add heat to

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the halon when pumping liquid, and shall add a maximum of 20 °F superheat to halon vapor, when pumping saturated halon. vapor at an inlet temperature of 100 'F with a compressed air pressure setting of 100 psig, and an ambient temperature of 100 °F. There shall be no heat transferred to the halon by the pneumatic motor during pumping operation. There shall be no halon contamination during pumping operation. Maximum leakage allowed is 25-ounce per year during pumping operation. Pump seals shall be compatible with halon. Pump metallic parts shall be corrosion resistant material or have corrosion resistant finish and be compatible with halon. Pneumatic drive motor seals shall be suitable for compressed air operation.

3.12 Filter. A filter shall be provided for the halon fluid during the recovery cycle. The filter shall be the replaceable core type. The filter housing components shall have a corrosion resistant finish and be compatible with halon. The filter element (s) shall be a blend of desiccants of the molded porous core type. It shall have a minimum total desiccant volume of 96 cubic inches, and a minimum surface filtering area of 128 square inches.

3.13 Compound gauge. The compound gauge shall show the pressure and vacuum reading~ as the unit is removing liquid halon and vapor from the fire extinguisher during the recovery cycle. The gauge shall have a vacuum range of O to 30 inches mercury and a positive pressure range of O to 300 psig min., 500 psig max. The compound gauge shall be connected so as to show the pressure in the sight glass/moisture indicator and filter drier assembly. The gauge shall meet the legibility requirements of MIL-STD-1472.

3.14 Inlet and outlet pressure gauges. The inlet gauge shall show the pressure in the hose from the container connected to the pump inlet. The gauge shall be connected between the outlet of the flow direction valve and the pressure regulator. The outlet gauge shall show the pressure in the hose to the container connected to the pump outlet. The gauge shall be connected between the direction control valve and the pump outlet. The inlet and outlet pressure gauges shall be glycerin-filled or equal, and have a snubber to reduce oscillation due to pressure pulses caused by the pump. The gauges shall have a range of O to 1500 psig. The gauges shall meet the legibility requirements of MIL-STD-1472.

3.15 Sight glass/moisture indicator. When the system is operating in either the recharging or recovery mode, halon shall be visible through a clear sight glass lens. There shall be a moisture indicator in the center of the sight glass. This moisture indicator shall change from green to yellow as the moisture content of the halon increases. A color guide shall be located above the sight glass for easy reference.

3.16 Strainer. The halon inlet line to the pump shall have a strainer to catch any contamination or particles that may be in the halon lines. The strainer shall be an 80-mesh screen.

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3.17 Nitrogen pressure regulator. The nitrogen pressure regulator shall be capable of reducing pressure of an initial source of 2,265 psi to a delivery pressure between O and 1,500 psi, with a deliver-y rate of not less than 25 cfm of nitrogen at any combination of inlet and outlet pressures within the design limits. The pressure-adjustment control shall consist of a screw provided with a suitable handle which shall require a maximum of 30 inch-pounds of torque to turn. The final delivery pressure shall not vary more than 3 percent from the preset value. A cushioned mounting bracket shall be provided to store the regulator when not in use. The mounting bracket shall prevent breakage and movement during shipment .

3.18 Halon and nitrogen cylinder rack. A stationary rack or straps to hold and secure the halon and nitrogen supply cylinders shall be fixed to the console at such a location as to allow simple and direct hookup between the cylinders and appropriate fittings and hoses . An adjustable device shall be furnished for rigidly securing the cylinders to the rack.

3.19 Dial indicator scale. The dial indicator scale shall consist of a dial face mounted on a pedestal which is attached to a horizontal base. The actuating mechanism for the scale must be located in the base on which the fire extinguisher cylinder is placed. The dial shall have a face not less than 13 inches in diameter and be capable of indicating weight from O to 100 pounds in 4 -ounce increments . The scale shall be equipped with a hand-operated adjuster that will permit the scale to be set at O pounds when 25 pounds is applied to the scale. The adjuster shall be an externally located knob or thumbscrew.

3.20 Thermometer . The thermometer shall be capable of measuring ambient temperatures from -20 to +120 ‘F and shall be attached to the rear of the scale pedestal . The thermometer shall have a clear, colorless plastic protective cover to prevent damage and should not be less than 11 inches high.

3.21 Extinguisher/containel mounting. Means shall be provided to hold the extinguishers/containers , delineated in Figure 3, being serviced in the proper position on the scale.

3.22 Safety goggles a n d gloves. A pair of safety goggles in accordance with MIL-STD-1434 and a pair of protective gloves in accordance with ZZ-G-381, type 111, style 1, size 10, 14 inches long, shall be provided for the operator.

3.23 Transportability. The unit shall be capable of being transported by all transportation modes, except airdrop and low altitude extraction, in accordance with MIL-STD-1366. The unit shall be capable of being transported on a 1-1/2 ton trailer. Allowable space for the halon recharger/recovery unit, including Halon 1301 cylinder and nitrogen cylinder, durinq transportion is 4 feet x 4 feet.

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3.24 Tiedown provisions. The tiedown provisions shall conform to MIL-STD-205, class 2 or 3, type IV equipment, and to MIL-STD-1791 for equipment restraint criteria. The tiedown provisions shall satisfactorily complete the pull testing as specified in 4.5.1, without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components. Tiedown points shall be labelled "TIEDOWN HERE" and combination lift/tiedown points shall be labelled "LIFT/TIEDOWN HERE". The marking shall conform to 3.27.

3.25 Slinging provisions. The slinging provisions shall conform to MIL-STD-209, class 1 or 3, type IV. The provisions shall enable the complete recharge/recovery unit to be lifted in the normal operating position. The provisions shall be located so that not less than 1-inch clearance is maintained between slings and all exterior parts, and shall be fastened to members which will withstand stresses in the amount and direction of pull specified for the provisions without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components. Slinging provisions may also be used as tiedown provisions when such provisions meet the requirements specified in 3.24. Lift points shall be labelled "LIFT HERE". The marking shall conform to the requirements of 3.27.

3.26 Stowage/hold-down provisions. Stowage and hold-down for hoses, tools, and adapters, shall be provided.

3.27 Human factors. The unit shall conform to the applicable human factors engineering criteria as specified in MIL-STD-1472 and shall be capable of operation by the 5th to 95th percentile.

3.28 Cover. A lightweight canvas cover shall be provided to protect the unit including supply cylinders from rain, snow, and dust during storage and transport.

3.29 Controls and gauges. All controls and gauges shall be identified with their function and positions for operation. Marking shall meet the legibility requirements of MIL-STD-1472. Controls and gauges shall be legible under red light and operable when wearing protective clothing.

3.30 Identification marking. The unit shall be identified in accordance with MIL-STD-130. Marking shall be applied to the unit with plates conforming to MIL-P-514, type I, style 1, composition C of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.31 Instruction plates. Each unit shall include an instruction plate or diagram, defining the recharging and recovery instructions for each of the fire extinguishers. Instructions shall include warnings and cautions, describing any special or important procedures to be followed in assembling, operating or servicing the unit. The instruction plate shall be 17-inch by 11-inch with a complete set of instructions printed in black characters on both

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sides of white cardboard stock in accordance with UU-C-190, grade A, type I, class 1, 5 plies thick. The instruction plate shall be laminated between two sheets of clear plastic in accordance with L-P-535, composition A, type I, class 3. The card shall be attached to the unit with a 24-inch metal chain.

3.32 Treatment and painting. All parts of the unit and its components normally painted shall be treated and painted in accordance with MIL-T-704, type F or G as applicable.

3.33 Marking. Mark tiedown and 'slinging points (see 3.24 and 3.25) in l-inch high characters, using white enamel, in accordance with TT-E-527, color no. 37875 of FED-STD-595,

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor's own facilities or any other, suitable for the performance of the inspection requirements specified herein, may be used. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system, or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract, Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material .

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

First article inspection (see 4.3.1) .
b Quality conformance inspection (see 4.4.2)
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

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4.3.2 Tests. The first article shall be tested *as* specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for inspection and tests shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4,5 Inspection procedures.

4.5.1 Examination. The unit shall be examined for the following defects:

101. Weight of units not as specified (see 3.2).
102. Material not as specified (see 3.4.1 and 3.4.2).
103. Safety not as specified (see 3.6).
104. Hoses, valves and fittings not as specified (see 3.7 and 3.8).
105. Air filter not as specified (see 3.9).
106. Air pressure regulator not as specified (see 3.10).
107. Pump not as specified (see 3.11).
108. Filter not as specified (see 3.12).
109. Compound gauge not as specified (see 3.13).
110. Pump discharge pressure gauge not as specified (see 3.14).
111. Sight glass/moisture indicator not as specified (see 3.15)
112. Strainer not as specified (see 3.16).
113. Nitrogen pressure regulator not as specified (see 3.17).
114. Halon and nitrogen cylinder rack not as specified (see 3.18)
115. Dial indicator scale not as specified (see 3.19).
116. Thermometer not as specified (see 3.20).
117. Safety chains no: as specified (see 3.21).
118. Safety goggles and gloves not as specified (see 3.22).
119. Transportability not as specified (see 3.23).
120. Tiedown provisions not as specified (see 3.24).
121. Slinging provisions not as specified (see 3.25).
122. Stowage/hold-down provisions not as specified (see 3.26).
123. Cover missing (see 3.28).
124. Controls and gauges not as specified (see 3.29).
125. Identification marking not as specified (see 3.30).
126. Instruction plates not as specified (see 3.31).
127. Treatment and painting not as specified (see 3.32).
128. Tiedown and lifting points marking illegible or missing (see 3.33).

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125. Dissimilar metals, as defined in MIL-STD-889, are not effectively insulated from each other (see 3.4.1 and 3.4.2)

4.5.2 Tests.

4.5.2.1 Test conditions. The unit is to be set up in accordance with the manufacturer's instructions. Examine the unit in regard to controls, safety and human factors.

4.5.2.2 Performance . The theory of operation of the recharge/recovery unit is to fill 5 7-, and 14-pound fire extinguisher cylinders with Halon 1 301 from a military 150 pound halon supply cylinder. The fire extinguisher cylinder is then overpressurized with nitrogen to approximately 750 psi at 70 'F. The unit shall be capable of recovering the halon/nitrogen mixture and transfer it to a holding cylinder. The unit shall be capable of drawing vacuum on the fire extinguisher. The recharge procedure must be accomplished in 10 minutes or less. No venting of halon to the atmosphere is permitted. The recovery procedure must be accomplished in 10 minutes or less. No venting of halon to the atmosphere is permitted.

4.5 .2.3 Test Procedure. The units shall be operated in accordance with manufacturer's instructions for recharging and recovering halon from 5-, 7-, and 14-pound fire extinguishers. When recharging , the extinguishers shall be overpressurized with nitrogen . Verify the following information:

a. The unit will perform the complete recharging operation in the time specified using the air compressor specified.

b. The unit will perform the complete recovery operation in the time specified using the air compressor specified.

4 .5.2.3.1 Halon transfer efficiency test,

a. All components which will contain Halon 1301 or nitrogen shall be weighed before testing and the weights recorded. This will include but not be limited to the supply and recovery and nitrogen cylinders, the pumping system, hoses, and the extinguishers. The tests shall be performed in a 70 'F nominal ambient temperature.

b. Five each 5-pound extinguisher cylinders shall be used for this efficiency test. The extinguishers shall be tested for leak tightness before start of test. The extinguishers shall be empty and at atmospheric pressure at start of test. Five extinguishers shall be charged with Halon 1301 and pressurized with nitrogen to a total pressure of 750 psig. The charging hose shall be disconnected after each charging cycle. The Halon 1301 and nitrogen shall be recovered from the extinguishers and placed in a recovery container. The valves on the extinguishers shall be opened for a minimum of

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5 seconds to confirm a vacuum was pulled after the halon has been recovered. The cycles of charging and recovery shall be done four times until a minimum of 100 pounds of halon, conforming to MIL-M-12218, has been charged into the extinguishers and then recovered. The total amount of halon transferred at this time shall be 200 pounds minimum.

c. The fire extinguishers shall now be charged with Halon 1301 from the recovery tank and pressurized with nitrogen/halon vapor from the vapor dome of the recovezy tank. The halon and nitrogen shall then be recovered back into the recovery tank as in (b) above. To accomplish charging and recovery the hose on the recovery tank must be moved from the liquid port to the vapor port ack as the procedure indicates. The hose to each extinguisher shall be disconnected and the valves opened for five seconds after each recovery. The cycle involving charging 5 extinguishers and recovering from 5 extinguishers shall be repeated for a total of 8 cycles. The total amount of used halon transferred in this test is 400 pounds.

d. A1l components shall be weighed as required in 4.5.2.3.l.a/ and the weight recorded. The difference in weight shall be considered as the Halon 1301 lost during the transfers.

e. The transfer efficiency shall be calculated as:

$$\frac{\text{Total halon transferred pounds minus halon lost pounds}}{\text{Total halon transferred pounds}} = \text{Efficiency}$$

The total halon transferred must exceed 600 pounds. The extinguisher charged weight minus the extinguisher empty weight times two equals the total halon weight transferred for an individual extinguisher charging and recovery cycle. These weoghts shall be recorded to determine the total Halon 1301 transferred during the test. The transfer efficiency must be 99 percent minimum.

4.5.2.4 Test objectives. The objectives of the tests are as follows:

a. Fill 5-, 7-, and 14-pound fire extinguishers with 1301 Halon and overpressurize with nitrogen.
b. Recover halon-nitrogen mixture to a holding tank.
c. Determine if the system can draw a minimum vacuum of 15 inches of mercury in the fire extinguisher cylinder.
d. Determine if there is any leakage of halon during charging, recovery cycles, connection/disconnect ion of hoses to extinguishers, supply bottles and holding tanks.
e. Determine if recharging cycle can be completed in 10 minutes or less.
f. Determine if the recovery cycle can be completed in 10 minutes or less.

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4.5.2 5 Failure criteria.

a. Halon 1301 transfer efficiency not as specified.
b. Nitrogen pressure cannot be adjusted by regulator.
c. Nitrogen fails to flow through any connection or hose assembly.
d. Any hose assembly or connection leaks.
e. Any component is deformed or damaged.
f. Recharge procedure not accomplished in 10 minnutes or less.
g. Recovery procedure nor accomplished in 10 minutes or less.
h. No venting of halon to the atmosphere is permitted during recharging or recovery.
i. Safety not as specified.
j. Human factors not as specified.
k. Controls and gauges not as specified.

4.6 Inspection of Packaging .

4.6.1 Quality conformance inspection of packing.

4.6.1.1 Unit of Product . For the purpose of inspection, a completed pack, prepared for shipment shall be considered a unit of product .

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects.

130. Disassembly not as specified (see 5.1)
131. preservative applied to surfaces of components that will contact Halon 1301 (see 5.2) .
132. Openings not sealed as specified (5.3) .
133. Unprotected surfaces not preserved as specified for level A (see 5.4.1.2).
134. Hoses and fittings not preserved in accordance with referenced document for level A (see 5.4.1.3) .
135. Blocking, bracing, and anchoring not in accordance with the referenced document for level A (see 5.5.1) .
136. Strapping not as specified for level A (see 5.5.1) .
137. Marking missing, illegible, incorrect, or incomplete (see 5.6).

5. PACKAGING

5.1 Disassembly. Disassembly shall be confined to the minimum necessary to effect a reduction in cube and safeguard parts vulnerable to damage if not otherwise removed. Disassembly shall also be confined to those parts that can be removed and reinstalled without special tools. Removed fasteners shall be reinstalled in mating parts and secured.

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5.2 Caution. Sufaces of components that will contact Halon 1301 shall not be coated with any preservatives.

5.3 Openings. All openings shall be sealed with plastic or metal plugs or with tape conforming to PPP-T-60, type IV.

5.4 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.4. Level A.

5.4.1.1 Preservatives. preservatives specified herein shall conform to the applicable specifications listed herein and shall be applied in accordance with MIL-P-116.

5.4.1.2 Unprotected surfaces. Unprotected exterior metal surfaces of components requiring the application of contact preservatives in accordance with MIL-P-116 and not specifically specified herein, shall be coated with type P-1 preservative.

5.4.1.3 Hoses and fittings. Hoses and fittings shall be preserved in accordance with MIL-H-775, level A.

5.4.1.4 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts.

5.4.1.5 Technical publications. Technical publications shall be preserved using method IC-1 or XC-3 of MIL-P-116.

5.4.1.6 Consolidation. The preserved hoses and fittings shall be placed in a close-fitting box conforming to PPP-B-636, W5C.

5.4.2 Commercial. The recharge/recovery unit shall be preserved in accordance with the requirements of ASTM D 3951.

5.5 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.5.1 Level A. Each complete unit shall be packed in a close-fitting box conforming to PPP-B-601, overseas type, style optional or PPP-B-621/ class 2/ style optional. Strapping shall conform to ASTM D 3953, type I or type 2, grade 2, size as applicable. Blocking, bracing and anchoring of the contents to prevent damage shall be in accordance with ASTM D 3951.

5.5.2 Level B. Each complete unit shall be packed as specified for level A in 5.5.1 except boxes shall be domestic type.

5.5.3 Commercial. Each complete unit shall be packed in accordance with the requirements of ASTM D 3951.

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5.6 Marking.

5.6.1 Military marking. Marking for levels A and B shall be in accordance with MIL-STD-129.

5.6.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 3951.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The recharge/recovery unit is intended to recharge 5-, 7-, and 14-pound fire extinguishers with Halon 1301 and overpressurize the extinguishers with nitrogen. The same unit shall recover halon from the fire extinguishers specified. The recharge/recovery unit can be used to fill 150-pound supply cylinders with Halon 1301 from a 2,000-pound drum.

6.2 Acquisition requirements. Acquisition documents should specify The following.

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Time frame for submission of the first article model (see 3.3).
d. When first article is not required (see 3.3).
e. Degree of preservation and packing required (see 5.4 and 5.5).

6.3 First article. When a first article inspection is required, the items) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Subject term (key word) listing.

Field
Halon
Overpressurizing

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6.5 Changes from Previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes .

Custodian:

Army - AV

Preparing activity:

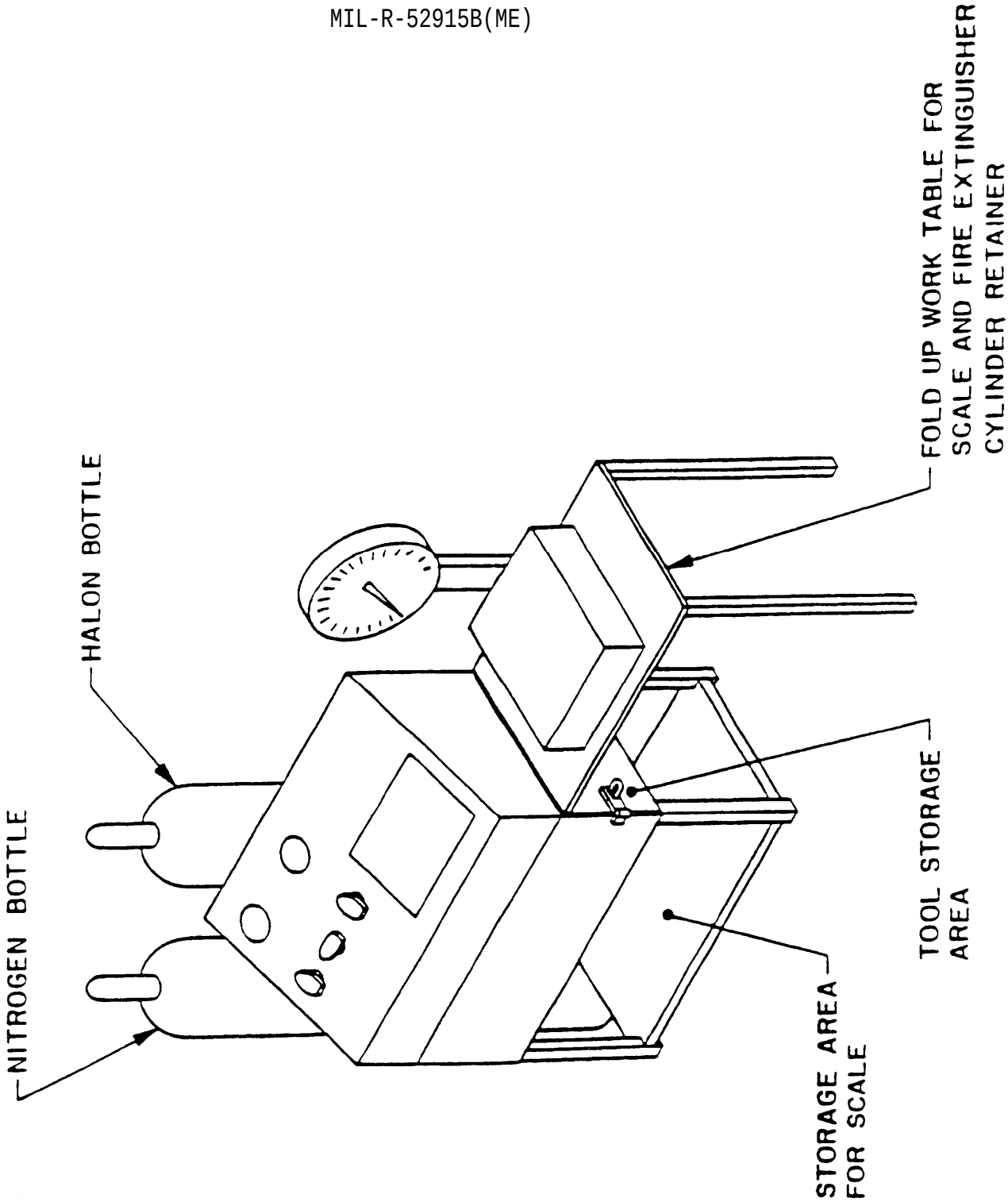
Army - AV

Agent activity:

Army - ME

Project 4210-A490

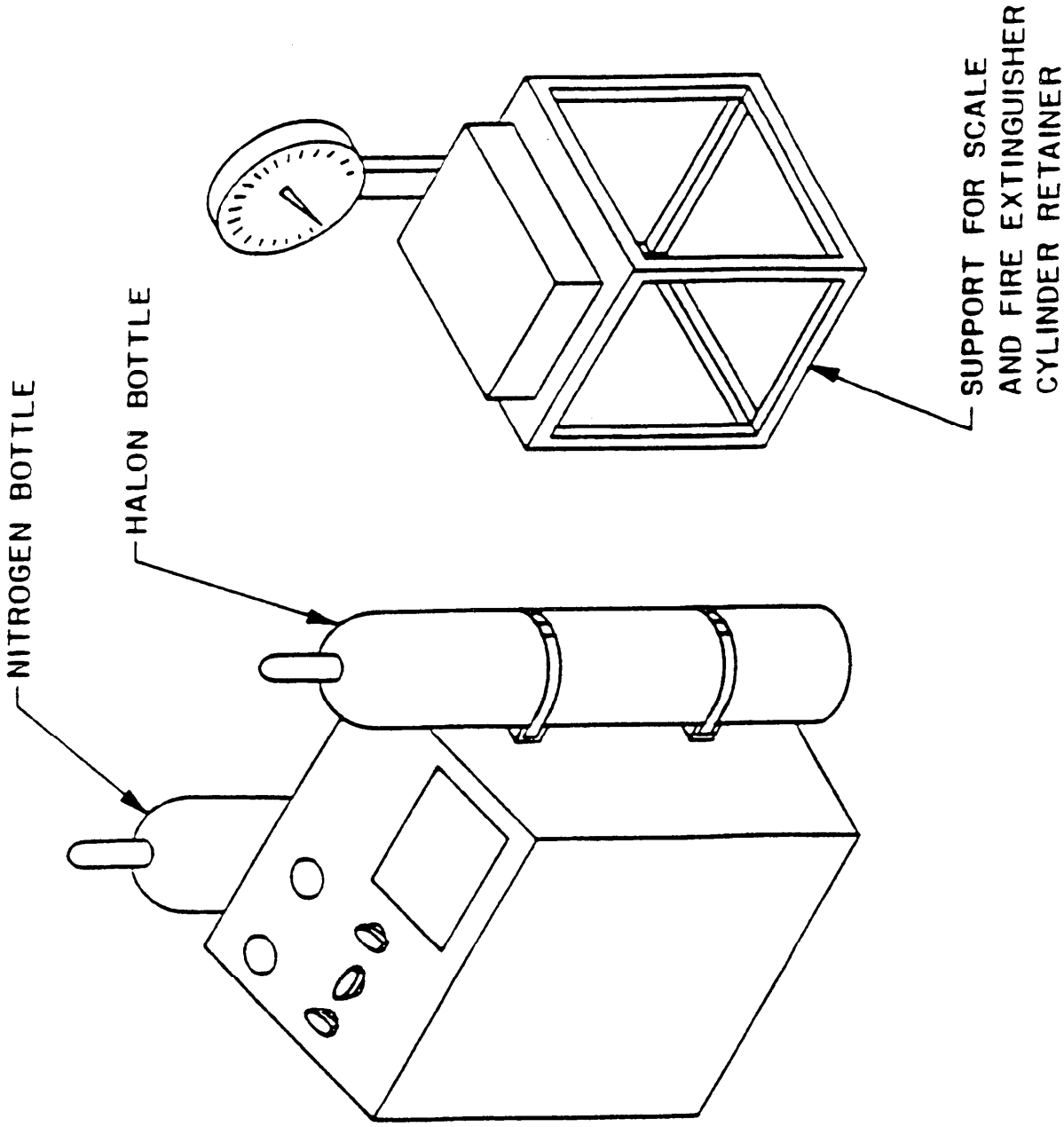
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FIGURE 1. RECHARGE/RECOVERY UNIT, FIRE EXTINGUISHER. MONOBROMOTRIRUOROMETHANE

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FIGURE 2. RECHARGE/RECOVERY UNIT, FIRE EXTINGUISHER, MONOBROMOTRIFLUOROMETHANE

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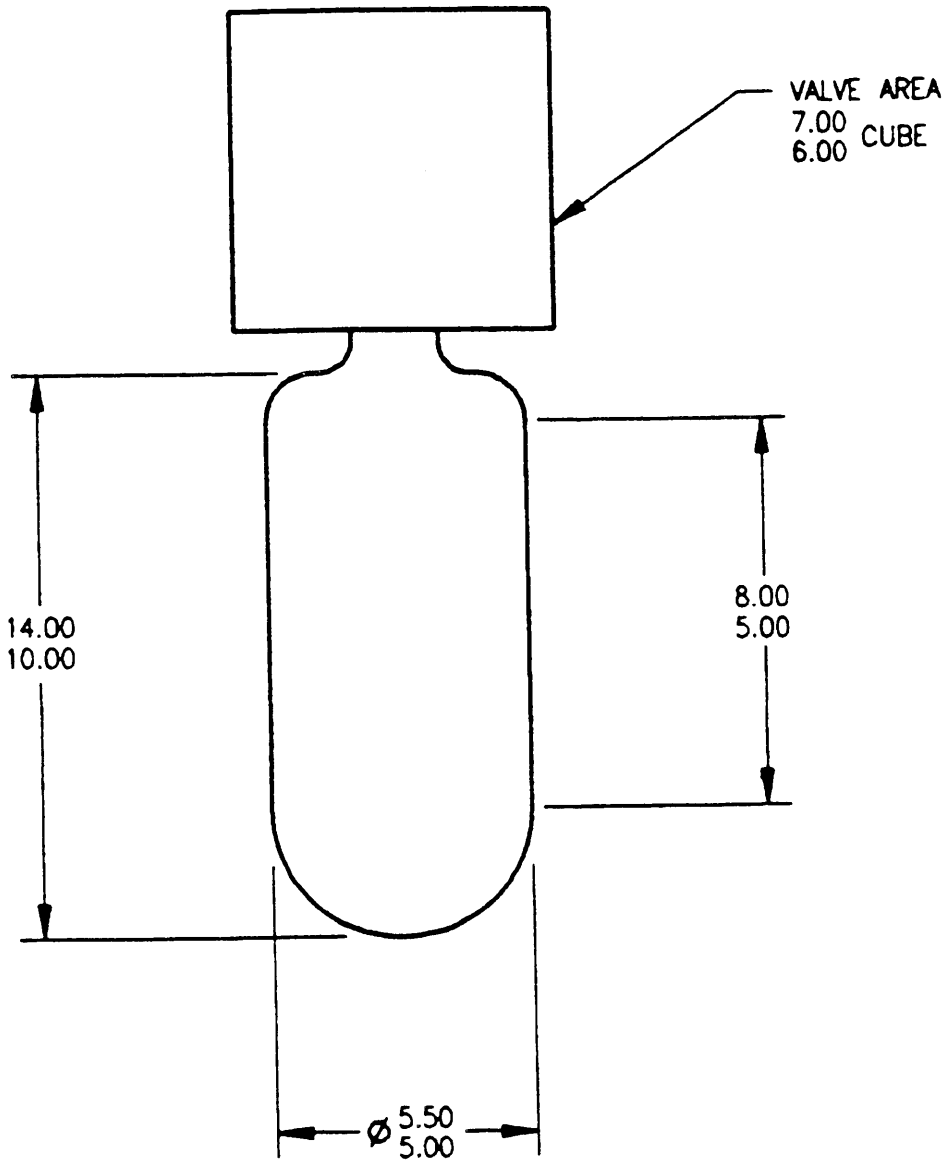


FIGURE 3. FIRE EXTINGUISHER ENVELOPE DIMENSIONS.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. submitter of this form must complete blocks 4, 5, 6, and 7.
3. preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-R-52915B(AV)	2. DOCUMENT DATE (YYMMDD) 940930

3. DOCUMENT TITLE
RECHARGE/RECOVERY UNIT, FIRE EXTINGUISHER MONOBROMOTRIFLUOROMETHANE (CF_3Br), PNEUMATIC POWERED

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) DSN	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) DSN
c. ADDRESS (Include Zip Code) MY AVIATION & TROOP CMD AMSAT-R-EDS 4300 GOODFELLOW BOULEVARD ST LOUIS MO 63120-1798	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: DEFNS QLTY & STDZN OFC 5203 LEESBURG PIKE STE 1403 FLS CHURCH VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

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